

1. Administrative & Study Identification

Full Study Title: A Phase I/IIa Randomized, Double-blind, Placebo-controlled Study to Evaluate the Safety and Pharmacokinetics of AZD7760 in Healthy Participants and in Patients with End-stage Kidney Disease Receiving Hemodialysis through a Central Venous Catheter **Short Title/Acronym:** PEAK Trial **Protocol Number:** D7480C00001 **NCT Number:** NCT06749457 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD7760 (mAb combination targeting *Staphylococcus aureus* virulence factors) **Phase of Development:** Phase I/IIa **Medical Monitor:** AstraZeneca Clinical Operations / PAREXEL

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and pharmacokinetics (PK) of AZD7760 when given as an intravenous infusion to healthy participants (Phase I) and to participants with end-stage kidney disease (ESKD) receiving hemodialysis through a central venous catheter (Phase IIa). **Secondary Objectives:** To evaluate the tolerability of multiple doses of AZD7760 compared to placebo, and to assess the immunogenicity (anti-drug antibodies) of the intervention.

2.2 Background & Rationale Patients with End-stage Kidney Disease (ESKD) who require hemodialysis through a central venous catheter are at an exceptionally high risk for life-threatening *Staphylococcus aureus* blood infections. This study tests AZD7760, a novel monoclonal antibody combination targeting *S. aureus* virulence factors. The early phase (I) ensures safety in healthy volunteers, while Phase IIa evaluates whether the prophylactic administration of the drug is safe and maintains adequate exposure levels in the vulnerable ESKD population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: 231 participants **Number of Sites:** Multi-center (Global/USA) **Estimated Study Start:** 30-Dec-2024 **Estimated Primary Completion:** 09-Oct-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. **Phase I:** Healthy male and female participants 18 to 55 years of age (inclusive); Body weight ≥ 45 kg and ≤ 110 kg with a BMI within 18.0 to 30.0 kg/m². 2. **Phase IIa:** Participants ≥ 18 years of age diagnosed with End-stage kidney disease (ESKD). 3. **Phase IIa:** Requiring hemodialysis through a tunneled central venous catheter as the primary vascular access for hemodialysis, having received hemodialysis for at least 90 days before randomization.

4.2 Exclusion Criteria

1. Known hypersensitivity to any component of the study intervention or severe adverse reaction following administration of monoclonal antibodies (mAbs). 2. Any condition that has the potential to increase clearance of the study intervention (e.g., protein loss conditions such as severe enteropathies, or plasmapheresis). 3. Receipt of immunosuppressive therapy, including any course of glucocorticoid therapy exceeding 2 weeks of prednisone (or equivalent >20 mg/day) within 6 months prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency:

* **Phase I:** Participants receive a single intravenous infusion of AZD7760 (escalating dose levels) or placebo. * **Phase IIa:** Participants are randomized to receive either AZD7760 or placebo administered as 2 intravenous infusions given 3 months apart (Day 1 and Day 91). **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** 3 months of active dosing (in Phase IIa), followed by a 12-month follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard ongoing hemodialysis therapies and standard-of-care medications for ESKD. **Prohibited:** Plasmapheresis or high-dose immunosuppressive therapies (e.g., prolonged systemic corticosteroids).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, ECG, Baseline Labs, Hemodialysis confirmation
Baseline (Phase I & IIa)	Day 1	Randomization, First IV Infusion, Safety & PK baseline blood draws	Interim (Phase IIa)
Day 91	Second IV Infusion (3 months post-initial dose), Safety monitoring, PK sampling	End of Study	Month 15 12-month Follow-up evaluation after the final Day 91 dose, Final AE/SAE assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability profile characterized by the incidence of treatment-emergent adverse events (TEAEs) and Serious Adverse Events (SAEs); along with comprehensive Pharmacokinetics (PK) profiling of AZD7760 exposure over time. **Measurement Method:** Clinical assessments, physical examinations, and Central Laboratory Analysis of serial serum PK samples.

7.2 Secondary & Exploratory Endpoints * Incidence of anti-drug antibodies (ADA) to assess the immunogenicity of AZD7760. * Exploratory assessments of *Staphylococcus aureus* infection rates and catheter-related bloodstream infections in the Phase IIa ESKD cohort.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical evaluation during the infusion days, regular follow-ups, and laboratory monitoring throughout the 12-month observation period. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for any life-threatening events or hospitalizations. **Data Monitoring Committee (DMC):** External independent safety review committee to evaluate dose-escalation criteria (Phase I) and ongoing safety (Phase IIa).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all participants receiving at least one dose of study drug) and PK Analysis Set (participants with measurable serum concentrations). **Sample Size Justification:** A target enrollment of $N = 231$ participants is estimated to provide sufficient power to evaluate safety, tolerability, and precise PK parameter curves across healthy and renally impaired populations. **Handling of Missing Data:** Standard handling utilizing mixed-effects models for repeated measures (MMRM) for PK and descriptive statistics for safety endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits managed by AstraZeneca and clinical research organization partners (e.g., PAREXEL). **Audit Trail:** Secure eCRF systems with automated data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: D7480C00001 **Registry Number:** NCT06749457 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PEAK Trial (AZD7760 for Chronic Kidney Failure) **Official Regulatory Title:** A Phase I/IIa Randomized, Double-blind, Placebo-controlled Study to Evaluate the Safety and Pharmacokinetics of AZD7760 in Healthy Participants and in Patients with End-stage Kidney Disease Receiving Hemodialysis through a Central Venous Catheter

12. Clinical Framework (Infection Prevention Specific)

Primary Condition: End-stage Kidney Disease (ESKD) & *Staphylococcus aureus* Infection Prevention **Diagnosis Threshold:** Patients requiring hemodialysis via a tunneled central venous catheter for ≥ 90 days **Medical Methodology:** Prophylactic Monoclonal Antibody (mAb) therapy targeting *S. aureus* virulence factors **Optimization Target:** Prevention of catheter-related *S. aureus* infections

13. Study Procedures & Methodology

Management Pathway: Specialized IV infusion scheduling aligned with regular hemodialysis sessions **Education Protocol (HCP):** Infusion administration, blind-maintenance procedures, and strict PK sample handling **Education Protocol (Patient):** Catheter site care, signs of localized/systemic infection, and reporting guidelines **Quality Audit Mechanism:** Strict monitoring of hemodialysis logs, infusion tolerances, and subsequent blood draws

14. Observation & Follow-Up Schedule

Total Duration: Up to 15 months (including a 12-month follow-up period post-last dose) **Onsite Visit Cadence:** Screening, Day 1 (Dose 1), Day 91 (Dose 2 for Phase IIa), and subsequent follow-up milestones **Remote Monitoring Frequency:** Routine safety calls interspaced with standard in-clinic hemodialysis appointments **Checklist Review Interval:** Concomitant with regular hemodialysis sessions

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	____	[DD-MMM-YYYY]				